

AC-QMS

Aditya Chemicals Quality Management System

Phase 1 — User Stories

Epic 1: Identity, Roles & Access Control

US-1: Super Admin accesses the entire system with no department restriction including user management, audit logs, and system configuration

US-2: QC Executive accesses Product Master creation, SPEC configuration, MOA review, AWS data entry, and document submission within the QC department

US-3: QC Manager accesses approval queues, batch creation, analyst assignment, and review of all QC Executive work within the QC department

US-4: QA Executive accesses QA-side independent work and views QC-approved documents in read-only mode within the QA department

US-5: QA Manager accesses final signature actions on every document and COA sign-and-issue authority within the QA department

US-6: System enforces five roles only (Super Admin, QC Executive, QC Manager, QA Executive, QA Manager) and does not allow creation of new roles in Phase 1

US-7: System enforces two departments only (QC and QA) in Phase 1 and applies department-specific color coding (QC green, QA blue)

US-8: System enforces the role-to-permission matrix at the data layer so that no user can call an API outside their permitted scope even if they bypass the user interface

US-9: System prevents the same user from performing two consecutive stages on any document so a creator cannot approve their own work and an approver cannot sign their own approval

US-10: System issues a short-lived access token of 15 minutes and a long-lived refresh token of 9 to 10 hours so analysts can work a full shift without re-login

US-11: System hashes every user password using bcrypt so plain-text passwords are never stored

US-12: System forces a password change on first login for every newly created user before any other action is allowed

US-13: System locks any account after 5 failed login attempts and keeps it locked for 15 minutes before unlocking

US-14: System requires fresh password re-entry for every approval, signature, and issuance action regardless of how long the user has been logged in

US-15: System captures the user's geolocation at login and stores it with the active session for audit traceability

US-16: System logs every role-based action across the entire batch lifecycle for compliance audit

Epic 2: User Management (Super Admin Console)

US-1: Super Admin creates a new user by entering full name, username, email, role, department, and initial password

US-2: System validates role-to-department mapping and prevents creation of invalid combinations

- Super Admin cannot be assigned a department
- QC Executive and QC Manager must belong to the QC department
- QA Executive and QA Manager must belong to the QA department

US-3: Super Admin toggles any user's status between Active, Inactive, and Locked at any time

US-4: Super Admin forces a password reset on any user account which requires the user to set a new password on next login

US-5: Super Admin generates a temporary password for any user and the user is required to change it on next login

US-6: Super Admin views the failed login attempt counter for every user and manually unlocks any locked account

US-7: Super Admin views the last login timestamp and last login IP address for every user

US-8: System logs every user-management action (create, modify, lock, unlock, reset) with the Super Admin's identity and timestamp

US-9: System prevents the Super Admin from deactivating their own account so the system always has at least one active admin

US-10: Bulk user import via CSV is not included in Phase 1 and is deferred to Phase 2

Epic 3: Product Master Foundation

US-1: System prevents creation of any SPEC, MOA, AWS, COA, or batch unless an approved Product Master exists for that product

US-2: QC Executive creates a Product Master by entering product identity, codes, regulatory references, and storage details

- Product name
- Product master family code (used for SPEC and MOA numbering)
- Product short code (used for AWS, COA, and batch numbering)
- Chemical name, chemical formula, molecular weight
- Regulatory references (IP, BP, USP, EP)
- Origin or source
- Method of production
- Composition
- Technical name

US-3: QC Executive enters the storage and shelf-life details on the Product Master

- Shelf life in months
- Storage conditions
- Special handling requirements
- Packing (primary and secondary)
- Sampling plan
- Retained sample quantity and retention period

US-4: System auto-calculates every batch's expiry date by adding the Master's shelf life to the manufacturing date

US-5: System assigns the initial revision number R-00 to every new Master and increments to R-01, R-02 on each subsequent revision

US-6: System preserves old revisions as Superseded and keeps old batches on their original revision so revision changes never affect existing batches

US-7: System captures revision history per Master with specification number, revision description, change control number, and effective date

US-8: QC Executive submits the Master and the QC Manager approves it before it can move to QA signature

US-9: QA Manager signs the Master after QC Manager approval and only then does the Master become live and usable for SPEC and MOA generation

US-10: System sets the document numbering format per product on the Master so every downstream document follows the configured format

Epic 4: Test Parameter Configuration (Field Master)

US-1: QC Executive defines test parameters under each Product Master with a sort order that controls how tests appear in SPEC, MOA, AWS, and COA

US-2: QC Executive classifies every test parameter as either Qualitative or Quantitative which controls how the test renders during data entry

US-3: System supports multiple field types for test parameters

- Qualitative text (e.g., Description, Solubility)
- Pass or Fail (e.g., Identification A, Identification B)
- Quantitative range (e.g., pH 5.9 to 6.3)
- Quantitative NMT or Not More Than (e.g., Heavy metals NMT 10 ppm)
- Quantitative NLT or Not Less Than (e.g., Assay NLT 98.5%)
- Quantitative BETWEEN (e.g., Assay 98.5% to 101.5%)

US-4: QC Executive configures each test parameter with name, sort order, result type, acceptance criteria text, min/max values, operator, unit of measurement, responsible department, mandatory flag, outside-lab flag, required instruments, and required reagents

US-5: System supports sub-tests under a parent test for cases like Identification A/B, Chlorides IP/BP/USP, and Assay IP/BP/USP with each variant carrying its own acceptance criteria and procedure

US-6: QC Executive applies a pharmacopoeia tag (IP, BP, USP, or EP) to each test variant to distinguish between different procedural references

US-7: QC Executive uses the Reset Right Side Limit button on quantitative test limits to clear the upper bound when only a lower bound is required

US-8: QC Executive views a live preview of how each test will render in SPEC and AWS forms before saving the Master

US-9: QC Executive defines a per-test calculation formula with named variables and the system reuses the same formula across MOA and AWS

US-10: System supports multi-step calculations where intermediate variables feed into the final formula (e.g., Loss on Drying feeds into the dried-basis Assay calculation)

US-11: QC Executive marks tests as customer-specific in the Customer Tab and configures customer-specific acceptance criteria overrides per test

Epic 5: MOA Sections (Method of Analysis Library)

US-1: QC Executive defines an MOA section under each test parameter on the Product Master so the procedure travels with the Master

US-2: QC Executive enters the per-section content covering all procedural details

- Pharmacopoeia reference (IP, BP, USP, or EP)
- Sample preparation procedure
- Standard preparation procedure
- Blank preparation procedure
- Reagent and solution preparation details
- Instrument parameters (column, wavelength, flow rate, temperature)
- System suitability requirements
- Sequence table for chromatographic methods
- Step-by-step procedure text
- Calculation formula reference
- Conclusion template (Satisfactory or Not satisfactory)

US-3: System supports multi-pharmacopoeia procedure variants under a single test so Chlorides can carry IP, BP, and USP variants together

US-4: QC Executive uses a rich-text procedure editor with support for tables to capture calibration curve standards and instrument parameter tables

US-5: QC Executive marks procedures as outside-lab tests and adds Report Number and Date of Analysis placeholders for those tests

US-6: QC Executive captures microbiological procedure details including media composition tables, sterilization parameters, and incubation conditions

US-7: System ties MOA sections to the Master so they are never created or edited as standalone documents

Epic 6: Instrument & Reagent Master (Reference Masters)

US-1: QC Executive adds new instruments to the Instrument Master with instrument code, name, department, calibration date, use-before date, and status

US-2: System tracks each instrument's status as Active, Expired, Under Maintenance, or Retired

US-3: System displays a warning when a QC Executive selects an instrument whose calibration has expired during AWS data entry

US-4: QC Executive acknowledges an expired-instrument warning with a written comment before continuing AWS data entry

US-5: QC Executive adds new reagents to the Reagent Master with name, lot number, concentration, preparation date, expiry date, supplier, department, and status

US-6: System tracks each reagent's status as Active, Expired, or Consumed

US-7: System displays a warning when a QC Executive selects a reagent that has expired during AWS data entry

US-8: QC Executive acknowledges an expired-reagent warning with a written comment before continuing AWS data entry

US-9: System flags reference standard reagents separately from regular reagents (e.g., Reference Standard Glycine)

US-10: QC Manager views a calibration-expiry dashboard listing instruments with calibration due within 30 days

US-11: System logs every add, modify, retire, and expired-acknowledgement action on instruments and reagents to the audit trail

Epic 7: Batch Initiation & ARN Auto-Generation

US-1: QC Manager creates a new batch from the dashboard by selecting a product from the list of QA-signed Masters

US-2: QC Manager enters the Batch Number manually exactly as received from the Production department (e.g., GCN/010326)

US-3: System auto-generates the ARN (Analytical Record Number) at batch creation and never allows manual edit

US-4: System uses the format AR-[FY year]-[3-digit sequential] for every ARN (e.g., AR-2026-001)

US-5: System resets the ARN sequence to 001 at the start of each financial year (April 1)

US-6: System guarantees ARN uniqueness using atomic increment so two batches created at the same time never receive the same ARN

US-7: QC Manager enters the Manufacturing Date as Month + Year only with no day component

US-8: System auto-calculates the Expiry Date using Manufacturing Date plus the Master's shelf life

US-9: QC Manager enters batch metadata including batch size, quantity sampled, customer name, customer reference, customer special instructions, test request sheet number, received date, and testing date

US-10: QC Manager assigns a specific QC Executive to the batch so only that analyst can perform the data entry

US-11: System keeps Batch Number and ARN as separate fields throughout the system and never conflates them in any screen, report, or document

US-12: System sets the batch status to Active on creation and tracks the current document phase (SPEC, MOA, AWS, COA, or Released)

US-13: System allows only SPEC creation at batch initiation and locks MOA, AWS, and COA creation until predecessor documents are QA-signed

US-14: System logs batch creation with the QC Manager's identity, timestamp, and geolocation

Epic 8: SPEC Document Workflow (Specification)

US-1: QC Executive creates the SPEC document for a batch through a manual guided flow and not through a direct template fetch

US-2: System auto-generates the SPEC document number using the format SPEC/[product family code]/[seq] (R-[revision])

US-3: System auto-fills the SPEC from the approved Product Master with product identity, mandatory tests, acceptance criteria, limits, units, storage details, packing, retained sample details, and revision history

US-4: QC Executive reviews the auto-filled fields and activates optional tests as required by the specific customer

US-5: QC Executive applies customer-specific data from the Customer Tab when the batch is for a customer with special requirements

US-6: QC Executive uses the Reset Right Side Limit button on quantitative tests during SPEC configuration when only a lower bound applies

US-7: System prevents QC Executive from editing mandatory test acceptance criteria since only a Master revision can change those values

US-8: QC Executive submits the SPEC and the system requires fresh password re-entry plus geolocation capture before accepting submission

US-9: QC Manager reviews the submitted SPEC and either approves it or rejects it back to draft with a mandatory rejection comment

US-10: QA Manager signs the QC-approved SPEC with fresh password re-entry and only then does the SPEC reach signed status

US-11: System sends notifications to the document creator and approver whenever a SPEC is rejected at any stage

US-12: System auto-triggers MOA creation the moment the SPEC reaches signed status

Epic 9: MOA Document Workflow (Method of Analysis – Auto-Fetched)

US-1: System auto-creates the MOA document the moment the SPEC reaches QA-signed status with no manual fetch action required

US-2: System never displays a Fetch MOA Template button anywhere in the system

US-3: System auto-generates the MOA document number using the format MOA/[product family code]/[seq] (R-[revision])

US-4: System pulls the MOA content from the MOA Sections defined on the Master for every test included in the approved SPEC

US-5: System inherits sub-tests, pharmacopoeia variants, calculation formulas, and Customer Tab data from the SPEC

US-6: QC Executive reviews the auto-fetched MOA content and submits it without editing the content directly

US-7: System directs the QC Executive to the Master if MOA content needs to be changed since procedure changes must happen at the Master level

US-8: QC Manager reviews the submitted MOA and either approves it or rejects it back to draft with a mandatory rejection comment

US-9: QA Manager signs the QC-approved MOA with fresh password re-entry and only then does the MOA reach signed status

US-10: System prevents MOA creation until the SPEC has reached QA-signed status (sequential lock)

US-11: System auto-unlocks AWS creation the moment the MOA reaches QA-signed status

Epic 10: AWS Document Workflow (Analytical Work Sheet – Per Batch)

US-1: QC Executive creates a fresh AWS document for each batch and never inherits AWS data from another batch

US-2: System auto-generates the AWS document number using the format AWS/[product short code]/[seq]

US-3: System prevents AWS creation until the MOA has reached QA-signed status (sequential lock)

US-4: QA Manager issues the blank AWS to the QC department via the Issuance Stamp gate before lab work can begin

US-5: System renders the AWS in the exact structure of the client's existing paper form including header block, batch metadata block, summary results table, and per-test detailed sections

US-6: QC Executive views per-test sections containing instrument used (with calibration date and use-before), reagents used (with prep date and use-before), procedure text from MOA, editable inline placeholder values, observation field, calculation block, acceptance criteria, conclusion, and remarks

US-7: QC Executive uses a split-pane data entry view with the AWS form on the left and the SPEC + MOA reference documents visible on the right

US-8: QC Executive toggles between SPEC and MOA on the right pane while continuing data entry on the left

US-9: QC Executive uploads supporting files to any AWS section including PDFs, images of chromatograms, IR spectra, microscopy images, and outside-lab reports

US-10: System runs live calculations on numeric inputs and auto-determines the conclusion against acceptance criteria

US-11: System auto-detects Out Of Specification results when a value violates acceptance criteria and highlights the affected section in red

US-12: QC Executive acknowledges every Out Of Specification result with a written comment before the section can be marked complete

US-13: System logs OOS acknowledgement with timestamp and geolocation in the audit trail

US-14: System tracks each AWS section status as Not Started, In Progress, or Completed and prevents AWS submission until all sections are complete

US-15: QC Executive submits the AWS and the system captures password re-entry plus geolocation before accepting submission

US-16: QC Manager reviews the submitted AWS and either approves it or rejects it back to draft with a mandatory rejection comment

US-17: QA Manager signs the QC-approved AWS with fresh password re-entry and only then does the AWS reach signed status

US-18: System auto-triggers COA generation the moment the AWS reaches signed status

Epic 11: COA Auto-Generation & Issuance (Certificate of Analysis)

US-1: System auto-generates the COA from the QA-signed AWS data and never allows manual drafting of COA content

US-2: System auto-generates the COA document number using the format COA/[product short code]/[batch number]

US-3: System pulls all AWS sections in sort order and creates COA result rows with test name, result, acceptance limits, and conclusion

US-4: System determines compliance from the test results

- All Satisfactory → "Complies with the IP specification"
- Any Not Satisfactory → "Does not comply"

US-5: System pre-populates the signature block with Prepared By (from AWS), Checked By (from QC Manager approval), and leaves Approved By for QA Manager signature

US-6: System renders the COA in the exact layout of the client's existing paper form

- Header with product name, date of issue, batch size, batch number, manufacturing date (month + year only), and expiration date
- ARN displayed prominently
- Body table with Sr. No, IP Monograph Analysis, Specification, and Observation columns
- Sub-rows for Identification A and B with separate specification and observation
- Remark line showing the compliance statement
- Storage condition line from the Master
- Signature block with Prepared By, Checked By, and Approved By columns

US-7: QA Manager reviews the auto-generated COA and signs and issues it with fresh password re-entry plus geolocation capture

US-8: System sets the batch status to Released the moment the COA is issued

US-9: System sends notifications to Production and Stores departments when a batch is released

US-10: QC Executive, QC Manager, and QA Manager navigate back from the COA to AWS, MOA, SPEC, and Master using inline hyperlinks

US-11: Any user exports the COA as PDF for sharing with external parties (customers, regulators)

US-12: System prevents any edit to the COA after it has been issued and requires a formal change control process for any correction

Epic 12: Sequential Document Lock & State Machine

US-1: System enforces the sequential document chain at the data layer so no UI workaround can bypass the lock

- SPEC creation requires a QA-signed Product Master
- MOA creation requires SPEC signed status
- AWS creation requires MOA signed status
- COA auto-generation requires AWS signed status

US-2: System applies a uniform state machine to SPEC, MOA, and AWS: Pending → Draft → Submitted → QC Approved → QA Signed

US-3: System applies a COA-specific state path: Auto Generated → Issued (by QA Manager sign-and-issue)

US-4: System tracks the current document phase per batch (SPEC, MOA, AWS, COA, or Released)

US-5: System validates every state transition against role permission, no-self-approval check, predecessor state, fresh password verification, and geolocation capture

US-6: System rejects any state transition that is not on the explicitly allowed transition list

US-7: System triggers automatic side effects on transitions

- SPEC reaching QA Signed auto-creates the MOA in draft state
- MOA reaching QA Signed auto-unlocks AWS creation
- AWS reaching QA Signed auto-generates the COA
- COA reaching Issued releases the batch and notifies downstream departments

US-8: System logs every state transition with from-state, to-state, user, role, timestamp, geolocation, and any rejection comment

US-9: System preserves the full history of rejected and resubmitted documents in the timeline view for that document

Epic 13: Notification Center

US-1: Every user views an in-app notification panel showing all events that require attention or are relevant to their role

US-2: System generates notifications for every significant event

- Approval pending (action required)
- Document approved (informational)
- Document rejected (with rejection comment)
- QA Manager signed (informational, to creator and approver)
- Batch assigned to QC Executive

- AWS issuance by QA Manager (work can begin)
- OOS detected on an AWS section
- Expired instrument or reagent used in AWS
- Batch released (to Production and Stores)
- Master revision superseded
- Calibration or reagent expiry warning (to QC Manager)

US-3: Each notification displays type, title, message, deep link, read/unread status, and creation timestamp

US-4: User clicks any notification to navigate directly to the relevant document or screen

US-5: System displays an unread badge on the header indicating the count of unread notifications

US-6: User marks all notifications as read in a single click

US-7: User filters notifications by category (Pending Action, Informational, System)

US-8: System persists notifications in the database so they survive logout and login

US-9: Email notifications are not included in Phase 1 and are deferred to Phase 2

Epic 14: Audit Log & Compliance Trail

US-1: System writes every significant action to an append-only audit log that cannot be edited or deleted by any user

US-2: System captures the following action categories in the audit log

- User login, logout, failed login, and account lock
- Password changes
- User created, modified, deactivated (Super Admin actions)
- Master created, submitted, approved, signed, rejected, superseded
- Test parameter added, modified, deactivated
- Instrument or reagent added, modified, retired, expired-acknowledged
- Batch created, assigned, released, abandoned
- Document created (SPEC, MOA, AWS, COA)
- Document submitted, approved, signed, rejected
- AWS section started, completed, OOS detected, OOS acknowledged
- File attachment uploaded or deleted
- COA auto-generated, signed, issued

US-3: System records every audit entry with the following fields

- Timestamp (UTC and Indian Standard Time display)
- User ID, user name (preserved for history), role, department
- Action verb
- Entity type, entity ID, document number
- Field changed, old value, new value
- Comment (rejection reason, OOS acknowledgement, expired-item acknowledgement)

- IP address
- Geolocation (latitude, longitude, accuracy)
- User-agent

US-4: Super Admin, QC Manager, and QA Manager view the audit log through a dedicated viewer

US-5: Authorized users filter the audit log by user, role, department, entity type, document number, date range, and action

US-6: Authorized users search the audit log by free text across action, comment, and field columns

US-7: Authorized users export the filtered audit log to CSV

US-8: Authorized users view a timeline of all events for a single document in chronological order

US-9: System prevents update or delete on audit records at the database layer by not granting UPDATE or DELETE permissions on the table

US-10: System retains audit records indefinitely in Phase 1 for ICH Q7 compliance

Epic 15: Geolocation Capture (Audit Reinforcement)

US-1: System captures the user's geolocation at the following actions

- Login
- Document submission
- Document approval
- Document signing
- COA issuance
- OOS acknowledgement
- Expired-item acknowledgement
- Master revision actions

US-2: System stores latitude, longitude, and accuracy in meters for every captured location

US-3: System prompts the user for geolocation permission on first login and stores the choice

US-4: System handles permission-denied cases based on configured policy (block action or proceed with a "Permission Denied" marker)

US-5: Authorized users view captured geolocation as part of the audit log entries

US-6: System does not track location continuously and captures only at the defined action points

US-7: Out-of-premises detection using geofencing is deferred to Phase 2

Epic 16: Session Management & Security

US-1: System provides every user with a session that lasts a full 9 to 10 hour work shift without expiring during the shift

US-2: System does not apply an idle timeout so analysts can step away from their terminal without being logged out

US-3: System tracks every active session with user, refresh token, expiration time, IP address, user-agent, and geolocation

US-4: User views their own active sessions across devices from their profile

US-5: System invalidates all refresh tokens for a user when that user changes their password (force logout on password change)

US-6: Super Admin force-logs-out any user by ending a selected session or all of that user's sessions

US-7: System requires fresh password re-entry for every approval and signature regardless of how long the session has been active

US-8: System allows a 60-second grace window for multiple sequential approvals so users do not re-enter the password for each approval within that window

US-9: System enforces HTTPS for all client-server communication

US-10: System applies CSRF protection on all state-changing endpoints using a synchronizer token

US-11: System rate-limits authentication endpoints to 5 attempts per minute per IP address

US-12: System stores the refresh token in a secure HTTP-only cookie and keeps the access token only in memory

Epic 17: Document Numbering & Format Configuration

US-1: System follows the client's existing paper-form numbering format for every document type generated

US-2: System applies the following numbering formats based on document type

- SPEC: SPEC/[family code]/[seq] (R-[rev]) → e.g., SPEC/FG00038/01 (R-02)
- MOA: MOA/[family code]/[seq] (R-[rev]) → e.g., MOA/FG00038/01 (R-02)
- AWS: AWS/[short code]/[seq] → e.g., AWS/GCN/01
- COA: COA/[short code]/[batch no] → e.g., COA/GCN/010326
- Batch Number: manually entered, typically [short code]/[DDMMYY] → e.g., GCN/010326
- ARN: AR-[FY year]-[3-digit seq] → e.g., AR-2026-001

US-3: QC Executive configures the numbering format per product on the Master using placeholder tokens (family_code, short_code, seq, rev, batch_no, fy_year)

US-4: System guarantees atomic sequence increment per document type and per financial year

US-5: System validates the format configuration at Master save and rejects formats that would produce duplicate document numbers

US-6: System prevents two batches of the same product from sharing a batch number

US-7: System resets sequences per financial year for ARN, per product family code for SPEC and MOA, and per product short code for AWS

Epic 18: Document Cross-Linking & Breadcrumb Navigation

US-1: User navigates from any document back to its parent documents using inline hyperlinks

US-2: User navigates from a COA to the source AWS, MOA, SPEC, Master (specific revision), and Batch

US-3: User navigates from an AWS to the reference SPEC, reference MOA, batch summary, instrument records used, and reagent records used

US-4: User navigates from a MOA to the SPEC and Master

US-5: User navigates from a SPEC to the Master and Batch

US-6: System displays breadcrumbs on every document page in the format: Home > Batches > [Batch No] > [Doc Type] > [Doc No]

US-7: User opens any cross-referenced parent document in a new tab to read without losing the current context

US-8: System opens parent documents in read-only mode when accessed through cross-links so the user cannot accidentally edit

Epic 19: Reports & Document Export

US-1: User exports any document (SPEC, MOA, AWS, COA) as PDF with a single click

US-2: System renders the exported PDF in the exact client paper format with same header, footer, columns, signature block, page numbers, and revision history table

US-3: System displays signature stamps in the exported PDF showing signer name, designation (Executive-QC, Manager-QC, Manager-QA), date, time, optional geolocation marker, and optional digital signature watermark

US-4: System applies a "DRAFT" watermark on unsigned documents and an "AUTO-GENERATED" watermark on unsigned COAs

US-5: QC Manager views a one-page batch summary report listing all four documents with status, document numbers, signatories, and timestamps

US-6: Authorized users generate a released-batches register filtered by date range, product, and customer

US-7: Authorized users generate an OOS register listing every Out Of Specification occurrence across batches with acknowledgement notes

US-8: Authorized users generate a Master register listing all products with their current revision and approval status

US-9: All reports export to PDF or CSV

US-10: Scheduled report emails and a custom report builder are not included in Phase 1 and are deferred to Phase 2

Epic 20: Database Schema & Persistence Layer

US-1: System uses PostgreSQL 16+ as the primary database with Prisma 5.x as the ORM

US-2: System organizes the 18-table schema into five logical groups

- Identity: users, departments
- Product Configuration: products, product_masters, test_parameters, moa_sections
- Batch Lifecycle: batches, batch_documents, aws_sections, coa_results, file_attachments

- Reference Masters: instruments, reagents
- System: audit_logs, notifications, sessions, arn_sequences

US-3: System keeps the audit_logs table append-only with no foreign key constraints so history is preserved even if referenced entities change

US-4: System uses JSONB fields for variable-shape data including formula variables, instruments required, reagents required, reagents used, and observations

US-5: System guarantees atomic sequence increment for ARN using the dedicated arn_sequences table with row-level locking

US-6: System applies soft-delete patterns for products, masters, and users using a status flag and never executes hard DELETE on those records

US-7: System allows hard delete only for unsaved draft attachments

US-8: System runs daily full backups and hourly incremental backups in Phase 1 operations

US-9: System creates indexes on batch_no, arn, doc_no, status fields, user/role lookups, and timestamp ranges on audit_logs

Epic 21: Tech Stack & Infrastructure

US-1: System backend runs on Node.js 20 LTS with Express.js 4.x and TypeScript 5.x

US-2: System uses Prisma 5.x as ORM, PostgreSQL 16+ as database, JWT for authentication, and bcrypt for password hashing

US-3: System stores files on local disk in Phase 1 and migrates to S3 in Phase 2

US-4: System frontend runs on React 18+ with Vite 5+ and TypeScript

US-5: System uses Tailwind CSS 3.x with shadcn/ui components, Zustand 4.x for state management, React Hook Form 7.x with Zod 3.x for forms, Axios for HTTP, and date-fns for date handling

US-6: System applies the brand color palette across the UI

- Primary deep teal #0D4F5C
- Accent medium teal #1A8FA3
- Highlight orange #E8732C
- Success #16A34A, Warning #C7780B, Danger #DC2626
- QC department green #2E7D32
- QA department blue #1565C0

US-7: System deploys on-premise at the client facility initially with cloud-ready architecture for Phase 2

US-8: System supports the latest two versions of Chrome, Edge, and Firefox

US-9: Mobile-responsive UI for tablets is not included in Phase 1 and is deferred to Phase 2

Epic 22: API Layer (REST Endpoints)

US-1: System exposes authentication endpoints for login, refresh, password verification (used for approval re-auth), password change, and logout

US-2: System exposes user management endpoints for Super Admin to list, create, view, modify, reset password, and unlock users

US-3: System exposes product and master endpoints to list products, create products, list masters per product, create new master revisions, view masters, edit drafts, and submit/approve/sign/reject masters

US-4: System exposes batch endpoints to list batches, create batches (QC Manager only), view batch details, and assign QC Executive

US-5: System exposes document creation endpoints

- POST /batches/:batchId/spec/create (QC Executive)
- POST /batches/:batchId/moa/create (internal call only, auto-triggered)
- POST /batches/:batchId/aws/create (QC Executive)
- POST /batches/:batchId/coa/generate (internal call only, auto-triggered)

US-6: System exposes document action endpoints for submit, approve, sign, reject, and sign-and-issue (COA only)

US-7: System exposes AWS section endpoints to list sections, view section, patch section, complete section, acknowledge OOS, acknowledge expired item, and upload attachments

US-8: System exposes reference master endpoints to list, create, and modify instruments and reagents

US-9: System exposes audit and notification endpoints

- GET /audit-logs (Super Admin, QC Manager, QA Manager only)
- GET /notifications
- PATCH /notifications/:id/read

US-10: System requires a valid JWT on every endpoint

US-11: System captures geolocation header on every state-changing endpoint

US-12: System requires fresh password verification on every approval endpoint

Epic 23: Dashboard & Pipeline Views

US-1: System renders a role-specific dashboard for each user on login

- Super Admin: user stats, login activity, audit-log summary, system health
- QC Executive: assigned batches, pending submissions, rejected drafts, notifications
- QC Manager: pending approvals, batches in progress, assign-analyst queue, calibration expiry alerts
- QA Executive: read-only view of QC documents, draft preparation area, QA-side independent work
- QA Manager: pending signatures, COAs awaiting issue, recently released batches, audit alerts

US-2: Authorized users view the batch pipeline in a Kanban-style layout with columns: New, SPEC in Progress, MOA in Progress, AWS in Progress, COA Pending, Released

US-3: System disables drag-and-drop on the pipeline so state changes happen only through legitimate document actions

US-4: Authorized users filter the pipeline by product, customer, date range, assigned analyst, and OOS-flagged

US-5: Authorized users click any batch card to open the batch dashboard with all four documents

US-6: User views the batch dashboard with a four-document tracker (SPEC, MOA, AWS, COA), three-stage approval indicator per document, quick links to parent documents, audit timeline, assigned analyst, batch metadata, and customer info

US-7: QC Manager views a calibration and expiry widget listing instruments with calibration due within 30 days

US-8: Every user views a pending-action widget on their dashboard with deep-links to the documents needing their action

Epic 24: Demo Data & Seeding

US-1: System loads seed data on first install for the prototype demo

US-2: System seeds five users with password "password123" and force-password-change enabled

- admin / Rajesh Kumar / Super Admin
- qc.exec / Kavya Patel / QC Executive / QC
- qc.mgr / Priya Mehta / QC Manager / QC
- qa.exec / Anand Joshi / QA Executive / QA
- qa.mgr / Sanjay Reddy / QA Manager / QA

US-3: System seeds the demo product Glycine IP with master family code FG00038, short code GCN, chemical formula $C_2H_5NO_2$, molecular weight 75.07 g/mol, regulatory references IP/BP/USP, shelf life 60 months (5 years), and storage in well-closed container at room temperature

US-4: System seeds the full test parameter set per the real client SPEC

- Description (Qualitative, mandatory)
- Solubility (Qualitative, mandatory)
- Identification A — By IR (Pass/Fail, mandatory)
- Identification B — By chemical test (Pass/Fail, mandatory)
- Appearance of solution (Pass/Fail, mandatory)
- pH (Quantitative range 5.9 to 6.3, mandatory)
- Chlorides (Quantitative NMT 100 ppm, mandatory)
- Heavy metals (Quantitative NMT 10 ppm, mandatory)
- Sulphated ash (Quantitative NMT 0.1%, mandatory)
- Loss on drying (Quantitative NMT 0.5%, mandatory)
- Assay on dried basis (Quantitative 98.5% to 101.5%, mandatory)
- Sieve analysis (Customer-optional)
- Bulk Density (Customer-optional)
- Elemental impurities by AAS — 10 metals per ICH Q3D (Customer-optional)
- Foreign matter (Qualitative, mandatory)

- Residual solvents — Methanol NMT 3000 ppm (Outside lab)
- Ethylene oxide NMT 0.1 ppm (Outside lab)
- Bacterial endotoxin (NMT 5 EU/g)
- TAMC (NMT 10^3 CFU/g)
- TYMC (NMT 10^2 CFU/g)
- E. coli, Salmonella, P. aeruginosa, S. aureus, Coliforms (Should be absent)

US-5: System seeds reference instruments: BAL-001 (Balance), HPLC-001, PH-001, OVEN-001, FTIR-001, AAS-001, GC-001

US-6: System seeds reference reagents: Mobile Phase Buffer, Reference Standard Glycine, Methanol HPLC Grade, Buffer pH 4.0, Buffer pH 7.0, 0.1 M Perchloric Acid, Crystal Violet Indicator

US-7: System seeds the demo batch GCN/010326 with ARN AR-2026-001, manufacturing date March 2026, expiry date February 2031, and batch size 3000 kg

US-8: System seeds demo test results matching the real client COA

- Description: A white, crystalline powder, odorless
- Solubility: Complies
- Identification A: Complies
- Identification B: Complies
- Sulphated ash: 0.04 %
- Chlorides: < 100 ppm
- Heavy metals: < 10 ppm
- Loss on drying: 0.22 %
- Appearance of solution: Complies
- pH: 6.01
- Assay on dried basis: 99.75 %

Epic 25: Open Items Awaiting Client Confirmation

US-1: Client confirms the exact document numbering format from the existing paper system for edge cases involving multiple products and multi-batch days

US-2: Client confirms that the SOP module is deferred to Phase 2 and is not part of Phase 1 scope

US-3: Client confirms that only QC and QA departments exist in Phase 1 and expansion logic for additional departments is deferred

US-4: Client confirms the geolocation consent flow — whether geolocation is opt-in or mandatory, and the system behavior when permission is denied

US-5: Client confirms that the 9 to 10 hour session has the idle timeout fully disabled

US-6: Client confirms whether additional sample products beyond Glycine IP should be seeded for the prototype demo

US-7: Client confirms the exact field list for the Customer Tab based on the client's existing customer-management practice

US-8: Client confirms whether SPEC and MOA should accept file attachments (Phase 1 currently allows attachments only on AWS)

US-9: Client confirms whether QA must formally issue blank AWS via the Issuance Stamp gate before QC Executive begins data entry, or whether this is purely a print-form artifact

US-10: Client confirms whether the COA signature block has 3 columns (per uploaded client COA) or 4 columns (per Development Bible) as the final layout

US-11: Client confirms whether the master family code (FG00038) and product short code (GCN) coexist on every product or whether one is derived from the other

US-12: Client confirms whether real AWS uses IP only or whether BP and USP pharmacopoeia variants run in parallel for export batches

Epic 26: Phase 2 Roadmap (Out of Scope for Phase 1)

US-1: SOP module with customizable headers and footers per department is deferred to Phase 2

US-2: "SOP of SOP" concept (master SOP governing SOP creation) is deferred to Phase 2

US-3: Email notifications via SMTP are deferred to Phase 2

US-4: S3-based file storage with CDN delivery is deferred to Phase 2

US-5: Mobile-responsive UI for tablets used on the shop floor is deferred to Phase 2

US-6: Scheduled report emails (daily released batches, weekly OOS register) are deferred to Phase 2

US-7: Custom report builder is deferred to Phase 2

US-8: Out-of-premises geofencing alerts are deferred to Phase 2

US-9: Multi-facility support (different plants, different document number sequences) is deferred to Phase 2

US-10: Bulk user import via CSV is deferred to Phase 2

US-11: API integration with ERP or SAP for batch number sync is deferred to Phase 2

US-12: Electronic-signature compliance with 21 CFR Part 11 (US FDA) for export markets is deferred to Phase 2

US-13: Document version control with diff viewer is deferred to Phase 2

US-14: Multi-language UI (English, Hindi, Gujarati) is deferred to Phase 2

US-15: Stability study module (long-term storage testing) is deferred to Phase 2

US-16: Change control module (linked to Master revisions) is deferred to Phase 2

US-17: CAPA (Corrective and Preventive Action) module is deferred to Phase 2

US-18: Deviation management module is deferred to Phase 2